AWG Wire Size Chart

Every standard size from 18 AWG to 1000 kcmil, with its diameter, its area in circular mils and square millimetres, and its resistance in copper and aluminum.

In the American Wire Gauge system a smaller number is a larger wire, and every three sizes the area doubles. A 10 AWG conductor has about twice the copper of a 13 AWG and half the resistance; go up six sizes and the diameter doubles. Above 4/0 the gauge numbers stop and sizes are given directly by area, in thousands of circular mils (kcmil).

AWG and kcmil wire sizes

Size	Diameter (in)	Diameter (mm)	Area (cmil)	Area (mm²)	Copper (Ω/1000 ft)	Aluminum (Ω/1000 ft)
18 AWG	0.0402	1.02	1,620	0.821	7.963	13.086
16 AWG	0.0508	1.29	2,580	1.31	5.000	8.217
14 AWG	0.0641	1.63	4,110	2.08	3.139	5.158
12 AWG	0.0808	2.05	6,530	3.31	1.975	3.247
10 AWG	0.1019	2.59	10,380	5.26	1.243	2.042
8 AWG	0.1285	3.26	16,510	8.37	0.781	1.284
6 AWG	0.1620	4.11	26,240	13.3	0.492	0.808
4 AWG	0.2043	5.19	41,740	21.1	0.309	0.508
3 AWG	0.2294	5.83	52,620	26.7	0.245	0.403
2 AWG	0.2576	6.54	66,360	33.6	0.194	0.319
1 AWG	0.2893	7.35	83,690	42.4	0.154	0.253
1/0 AWG	0.3250	8.25	105,600	53.5	0.1222	0.2008
2/0 AWG	0.3648	9.27	133,100	67.4	0.0969	0.1593
3/0 AWG	0.4096	10.40	167,800	85.0	0.0769	0.1263
4/0 AWG	0.4600	11.68	211,600	107	0.0610	0.1002
250 kcmil	0.5000	12.70	250,000	127	0.0516	0.0848
300 kcmil	0.5477	13.91	300,000	152	0.0430	0.0707
350 kcmil	0.5916	15.03	350,000	177	0.0369	0.0606
400 kcmil	0.6325	16.06	400,000	203	0.0323	0.0530
500 kcmil	0.7071	17.96	500,000	253	0.0258	0.0424
600 kcmil	0.7746	19.67	600,000	304	0.0215	0.0353
750 kcmil	0.8660	22.00	750,000	380	0.0172	0.0283
1000 kcmil	1.0000	25.40	1,000,000	507	0.0129	0.0212

Diameter is that of a solid round conductor of the stated area. A stranded conductor of the same size is slightly larger overall. Resistance is DC, at about 75°C.

How the values are calculated

$$\text{Area (cmil)} = (\text{diameter in mils})^2$$

$$\text{Area (mm}^2\text{)} = \text{cmil} \times 0.0005067$$

$$R\ (\Omega/1000\ \text{ft}) = K \times 1000 / \text{cmil}$$

Every figure in the chart comes from the formulas above. Nothing is copied from a published table, so any cell can be checked by hand.

What this chart does not tell you

This chart gives geometry and resistance. It does not give ampacity, which depends on insulation temperature rating, how many conductors share a raceway, the ambient temperature and the terminations, and has to come from the adopted code. Resistance is DC; on large conductors carrying AC, skin effect raises the effective figure.

This is a reference, not a design. It is here to give you the right order of magnitude and a figure to check against. It does not replace the adopted code or a calculation by a qualified engineer.

From voltfield.org/charts/awg-wire-size-chart.html, where the notes on reading this chart and a calculator for other cases are. Reviewed November 11, 2026. Free to print, copy and share.